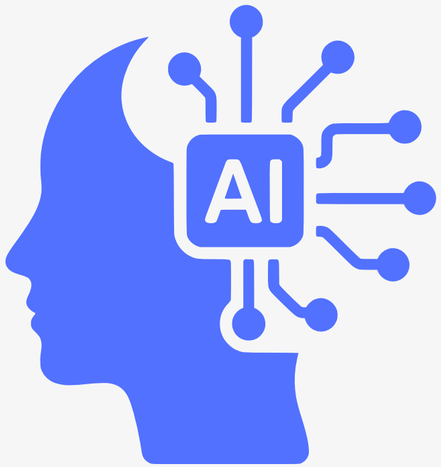
TechZeen



MCP & SECURE TOOL ECOSYSTEMS

IMAGINE YOUR AI ASSISTANT COULD READ FILES, QUERY DATABASES, ACCESS APIS, AND INTERACT WITH REAL TOOLS INSTEAD OF JUST GENERATING TEXT.

TechZeen

MODEL CONTEXT PROTOCOL (MCP)

- MCP: An open protocol that helps AI applications connect to external tools, data, and resources through a standardized interface.
- Why MCP? Without a common protocol, developers often need custom integrations for different tools.
- Example: An AI assistant uses an MCP server to access project files.

TechZeen

MCP ARCHITECTURE

- MCP Host: The AI application the user interacts with.
- MCP Client: Manages the connection between the host and an MCP server.
- MCP Server: Exposes specific tools, resources, or prompts.
- Tools: Actions the AI can request, such as searching files or querying a database.
- Resources: Data the AI can access, such as documents or project information.

TechZeen

FUTURE OF AI TOOLING

- Standardized integrations can reduce custom development work.
- AI agents can connect to multiple external systems.
- Tools can make AI applications more useful and capable.
- Security, permissions, and human approval remain essential.

TechZeen

SECURE MCP SERVERS

- Permission systems: Control which tools and data an AI application can access.
- Local tools: Tools running on your own computer or local environment.
- Sandboxing: Restrict a tool to a controlled environment to reduce potential damage.
- Authentication: Verify who is connecting to a server.
- Input validation: Validate tool arguments before execution.
- Least privilege: Give each tool only the permissions it needs.
- Human approval: Require confirmation for sensitive actions.